

Data Warehouse Projects

Project Requirements

Objective

Develop a modern data warehouse using SQL Server to consolidate sales data, enabling analytical reporting and informed decision-making.

Specifications

- Data Sources: Import data from two source systems (ERP and CRM) provided as CSV files.
- Data Quality: Cleanse and resolve data quality issues prior to analysis.
- Integration: Combine both sources into a single, user-friendly data model designed for analytical queries.
- Scope: Focus on the latest dataset only; historization of data is not required.
- Documentation: Provide clear documentation of the data model to support both business stakeholders and analytics teams.

Project Approach

In the projects, the team have decided to build data warehouse with Medallion Architecture Approach due to:

- Separation of concerns: each layer has one job, which keeps the code organized and easier to maintain.
- Reprocessing is simple: if cleaning rules change, we may rebuild Silver and Gold from Bronze.
- Easier teamwork and testing: we can validate data quality at each layer before moving on.

	Bronze Layer	Silver Layer	Gold Layer
Definition	Store Original Raw Data	Clean And Transform Data	Business Value Data
Object Type	Tables	Tables	View
Load Method	Full Load	Full Load	
Data Transformation Steps	None	Data Cleaning Data Standardization Data Normalization Derived Columns Data Enrichment	Data Integration Data Aggregation Business Logic
Data Modelling	None	None	Start Schema Aggregate
Target Audience	Data Engineer	Data Engineer Data Analyst	Data Analyst Business Users